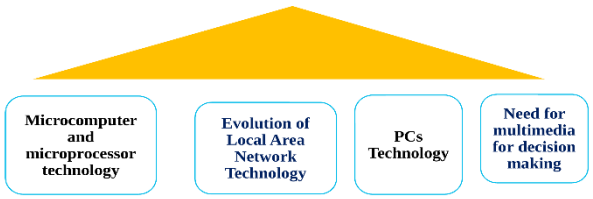
MAIN: ASS 2

NAME	مصطفى محمد السعيد محمد البكش
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CODE SUBJECT:	CMPn421
SUBJECT:	Distributed Computer Systems
SPECIALIZATION:	Computer engineering

ASS#2_DCS FALL 2026

1. Explain the technologies that move the computer systems from centralized to distributed.

A:



1. Advances in Microcomputer and microprocessor technology: The first was the development of powerful microprocessors (8-bit, then 16-, 32- and 64-bit CPUs, and multicore CPUs). Current machines have the computing power of the mainframes deployed 30 or 40 years ago, but for 1/1000th of the price or less.

2. Evolution of Local Area Network Technology: The second development was the invention of high-speed LAN and WAN computer networks. LANs allow thousands of machines within a building or campus to be connected, and WANs allow hundreds of millions of machines all over the earth to be connected.

3. PCs Technology: Miniaturization of computer systems, with the smartphone as the most impressive outcome, plus plug computers and Nano computers that offer near-desktop performance.

4. Need for multimedia for decision making: Decision makers are asking for multimedia (text, audio, video, graphs) for decision making. This requirement replaced text terminals with PCs and the use of powerful servers

The technologies that move the computer system systems from centralized to distributed

• The result of these technologies put together a computing system composed of many networked computers, be they large or small.

• These computers are generally geographically dispersed, for which reason they are usually said to form a distributed system.

• The size of a DCS vary from a handful of devices, to millions of computers.

• The interconnection network may be wired, wireless, or a combination of both.

• DCS are highly dynamic, in the sense that computers can join and leave, with the topology and performance of the underlying network almost continuously changing

Summary for IT Motivations for Distributed Systems

Feature	***Notes***
Technological Changes	Growth in microelectronics (e.g., VLSI – Very Large Scale Integration, etc) has enabled price-performance ratio to favor multiple low-performance processors rather than single high-performance processors. Interconnection and communication costs have fallen dramatically – LANs, etc are now readily available and cost-effective. PCs and mobiles are powerful, low cost devices, multifunction's, to replace terminals
User Needs	Increased desire to provide computing power (processing) where it is needed. Users need more sophisticated multimedia reports, facilities – faster, reliable, etc. There is also managerial desire to Distribute the Computing. Managers need multimedia data for better decisions-making

2. Mention the roles of microprocessor/minicomputer technology in the evolution of DCS?

A:

• The first was the development of powerful microprocessors.

• Initially, these were 8-bit machines, but soon 16-, 32-, and 64-bit CPUs became common. With multicore CPUs, we now are facing the challenge of adapting and developing programs to exploit clustering and parallelism.

• Current generation of machines have the computing power of the mainframes deployed 30 or 40 years ago, but for 1/1000th of the price or less.

• **Ex1: HP PC:** Windows 11, Up to 12 Core i9 processor, 14 threads, 12MB L3CACHE, Up to 128 GB DDR5 RAM, and 2.5 TB STORAGE

• **Ex2: Hyper-converged Infrastructure HCI** is a SDH (hardware/software-defined IT infrastructure) that virtualizes all of the elements of conventional "hardware-defined by software" systems. HCI includes clustering and virtualized computing

Hyperflex Hyper Converted Infrastructure HCI

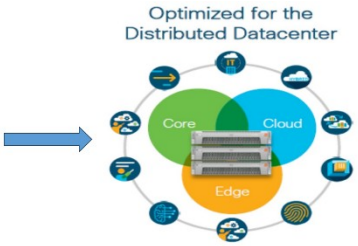
• Hyper-converged infrastructure HCI is a hardware/software-defined IT infrastructure that virtualizes all of the elements of conventional "hardware-defined" systems.

• HCI includes, at a minimum, clustered of at least 3 processors with virtualized computing…

• Hyper Converged Infrastructure. HCI Nodes All-Flash HX240C. 16 CPU, 512 cores, 1024 threats. 3 TB Memory, 200 TB SSD Storage



- **Hyper Converged Infrastructure.**
- HCI Nodes All-Flash HX240C.
  - 16 CPU
  - 512 cores, 1024 threats.
  - 3 TB Memory
  - 200 TB SSD Storage



The categories of computers and their uses based on differences in size, speed, processing capabilities, and price

• **personal computers:** A PC can perform all of its input, processing, output, and storage activities by itself. PCs include desktop computers and notebook computers.

• **handheld computers:** A handheld computer is a small computer that fits in hand. Handheld computers can perform specific, industry-related functions, or can be general-purpose. A PDA is a handheld computer that provides personal organizer functions, such as a calendar, appointment book, and notepad.

• **Internet appliances:** An Internet appliance is a computer with limited functionality whose main purpose is to connect to the Internet from home.

• **mid-range servers:** A mid-range server is more powerful and larger than a workstation computer. Users typically access a mid-range server through a personal computer or a terminal, which is a device with a monitor and a keyboard that usually has no stand-alone processing power.

• **mainframes:** A mainframe is a large, expensive, very powerful computer that can handle hundreds or thousands of connected users simultaneously.

• **supercomputers:** A supercomputer is the fastest, most powerful, and most expensive category of computers.

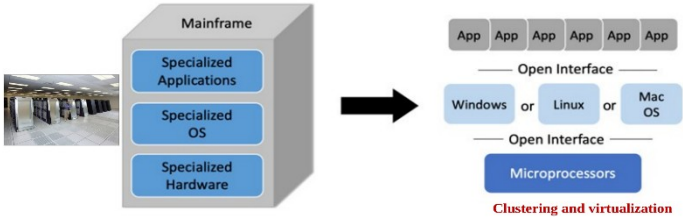
1. Advances in Microcomputer and microprocessor technology

The categories of computers and their uses based on differences in size, speed, processing capabilities, and price

personal computers	handheld computers	Internet appliances	mid-range servers	mainframes	supercomputers
A PC can perform all of its input, processing, output, and storage activities by itself. PCs include desktop computers and notebook computers.	A **handheld computer** is a small computer that fits in hand. Handheld computers can perform specific, industry-related functions, or can be general-purpose. A PDA is a handheld computer that provides personal organizer functions, such as a calendar, appointment book, and notepad.	An Internet appliance is a computer with limited functionality whose main purpose is to connect to the Internet from home.	A **mid-range server** is more powerful and larger than a workstation computer. Users typically access a mid-range server through a personal computer or a terminal, which is a device with a monitor and a keyboard that usually has no stand-alone processing power.	A **mainframe is a large**, expensive, very powerful computer that can handle hundreds or thousands of connected users simultaneously.	A **supercomputer** is the fastest, most powerful, and most expensive category of computer.

• Transformation of the vertical mainframe market to a horizontal marketplace with open interfaces and multiple options available at every level.

• The introduction of powerful microprocessors (e.g., Intel x86 and Motorola 68000) and open source OS's (e.g., BSD Unix and Linux), helped transform that vertical market into a horizontal marketplace, with open interfaces spurring innovation at every level



3. Mention the roles of local area and wide area networks technology in the evolution of DCS?

A: • The second development was the invention of high-speed LAN and WAN computer networks.

• LANs allow thousands of machines within a building or campus to be connected in such a way that small amounts of information can be transferred in a few microseconds or so. Larger amounts of data can be moved between machines at rates of 10^{12} bps.

• WANs allow hundreds of millions of machines all over the earth to be connected at speeds varying from 10^4 to 10^9 bps, and sometimes even faster.

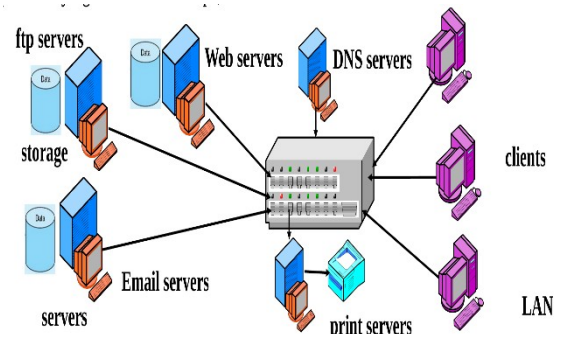
• A LAN consists of a number of different computers. Workstations WSs and PCs provide the front-end for network users. Different servers provide shared services.

• One or several network file servers provide data storage services. Any WS and PC may store files on disks maintained by these file servers.

• A local name server maps machine names to IP addresses, user names to user ids and group names to group ids. Any machine can request a service to resolve a certain name.

• One or several print servers control the access to shared printers. WSs and PCs have the server printing jobs for them.

• Another component provides a gateway to the WAN.



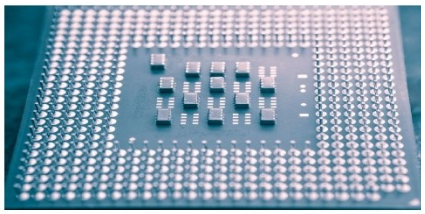
• Name servers provide the information regarding machine (by IP address), user names (by user ids), and group name (by group ids).

• LANs can be connected together to form Intranets.

4. Mention the roles of the PC technology in the evolution of DCS?

A:





• Parallel to the development of increasingly powerful, reducing costs, and networked machines, we have also been able to witness miniaturization of computer systems with perhaps the smartphone as the most impressive outcome. Packed with powerful PCs, sensors, lots of memory, and a powerful CPU, these devices are nothing less than full-fledged computers.

• Along the same lines, plug computers and other so-called Nano computers are finding their way to the market.

• These small computers, often the size of a power adapter, can often be plugged directly into an outlet and offer near-desktop performance.

• A PC desktop computers and notebook can perform all of its input, processing, output, and storage activities by itself.

• PCs and mobiles are powerful, low cost devices, multifunction's, to replace terminals

• decision makers are asking for multimedia (text, audio, video, graphs) for decision making. This requirement replace of text terminals with PCs and use of powerful servers.

Component	***Specification***
1. Processor (CPU)	Model: Intel Core i5-11400; Cores: 6; Threads: 12; Base Clock Speed: 2.6 GHz; Max Turbo Boost Speed: 4.4 GHz; Cache: 12MB Smart Cache
2. Memory (RAM)	Capacity: 16GB DDR4; Type: DDR4; Speed: 3200 MHz
3. Storage	Primary Storage: 512GB NVMe SSD; Secondary Storage: 1TB HDD (for additional storage)
4. Graphics Card (GPU)	Integrated Graphics: Intel UHD Graphics 730
5. Motherboard	Model: ASUS Prime B560M-A; Chipset: Intel B560; Form Factor: Micro ATX
6. Power Supply Unit (PSU)	Wattage: 500W; Efficiency Rating: 80 PLUS Bronze
7. Operating System	Windows 10 Pro
8. Connectivity	Ethernet: Gigabit Ethernet; USB Ports: 6 x USB 3.2 Gen 1, 4 x USB 2.0; Wi-Fi: Dual-band Wi-Fi 6 (802.11ax); Bluetooth: Bluetooth 5.1
9. Case	Model: Cooler Master Box Q300L; Form Factor: Micro ATX; Cooling: 2 x 120mm fans included
10. Monitor	Size: 24 inches; Resolution: Full HD (1920x1080); Refresh Rate: 60Hz

5. Mention the different applications/configurations of the PC in the evolution of DCS?

A:

• A PC desktop computers and notebook can perform all of its input, processing, output, and storage activities by itself.

• A desktop computer is designed so the system unit, input devices, output devices, and any other devices fit entirely on or under a desk or table. Variations of desktop computers include :

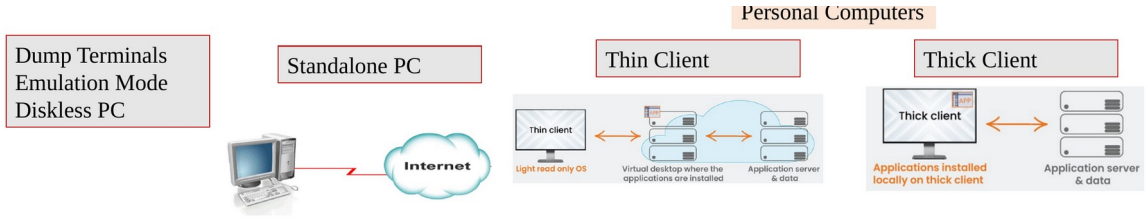
– **Tower models** (tall computers that can sit vertically on the floor),

– **All-in-one computers** (computers that combine monitor and system unit into a single device),

– **Workstations** (more expensive and powerful computers designed for work that requires intense calculation and graphics capabilities).

• A notebook computer is a portable PC small enough fit on lap.

• Notebook and desktop computers are used at home or in the office to perform application software-related tasks or to access the Internet.



• **handheld computers:** A handheld computer is a small computer that fits in hand. Handheld computers can perform specific, industry-related functions, or can be general-purpose. A PDA is a handheld computer that provides personal organizer functions, such as a calendar, appointment book, and notepad.

• **Internet appliances:** An Internet appliance is a computer with limited functionality whose main purpose is to connect to the Internet from home.

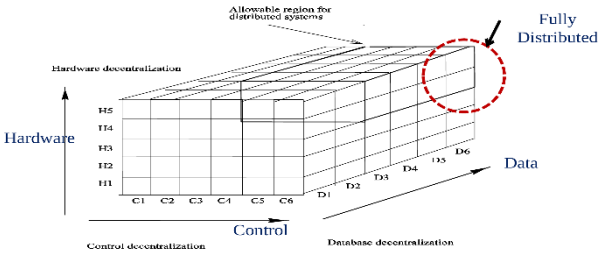
6. Draw and explain Enslow model for distributed processing systems

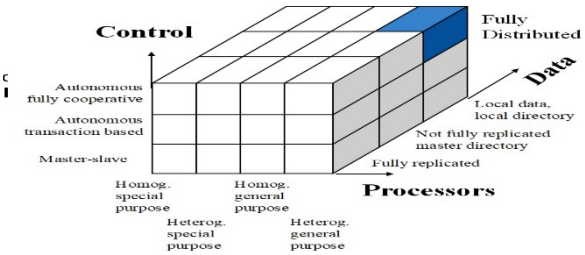
A:

A Distributed System = distributed hardware and/or distributed control and/or distributed data

• Distributed system = distributed hardware + distributed control + distributed data

• Thus, a system could be classified as a distributed system if all three categories (hardware, control, data) reach a certain degree of decentralization.





CONTROL

- Single fixed control point.
- Single dynamic control point. In multiple CPU cases the controller changes from time to time among CPUs.
- A fixed master/slave structure. For example, in a system with one CPU and one co-processor, the CPU is a fixed master and the co-processor is a fixed slave.
- A dynamic master/slave structure. The role of master/slave is modifiable by software.
- Multiple homogeneous control points where copies of the same controller are used.
- Multiple heterogeneous control points where different controllers are used.

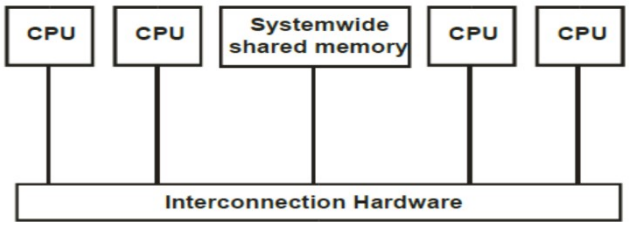
DATA

- Centralized databases with a single copy of both files and directory.
- Distributed files with a single centralized directory and no local directory.
- Replicated database with a copy of files and a directory at each site.
- Partitioned database with a master that keeps a complete duplicate copy of all files.
- Partitioned database with a master that keeps only a complete directory.
- Partitioned database with no master file or directory.

7. Explain Tightly Coupled Multiprocessor System?

A: • Tightly coupled systems are multiprocessor systems. They are a population of processors sharing addresses to memory and devices. This sharing is accomplished by utilizing single OS.

- Tightly coupled systems are referred to as parallel processing systems or shared memory, were a single system wide primary memory (address space) is shared by all the processors.
- If any processor writes, for example, the value 100 to the memory location x, any other processor subsequently reading from location x will get the value 100.

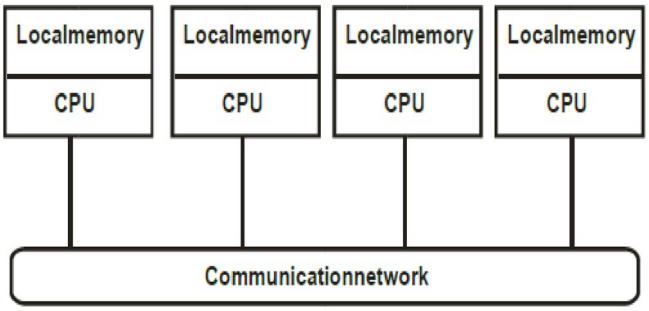


- In these systems, any communication between the processors takes place by the shared memory.

• In tightly coupled systems, the number of processors that can be deployed is small and limited by the bandwidth of the shared memory

8. Explain Loosely Coupled Multiprocessor System?

A:



• The loose interconnection of computer systems comes very close to the concept of distributed processing by the attribute of multiple complete systems.

• They are systems connected over I/O pathways traditionally distinct from networks.

• Many of these pathways differ from tightly coupled systems only in the physical sense that they do not share memory or operating systems.

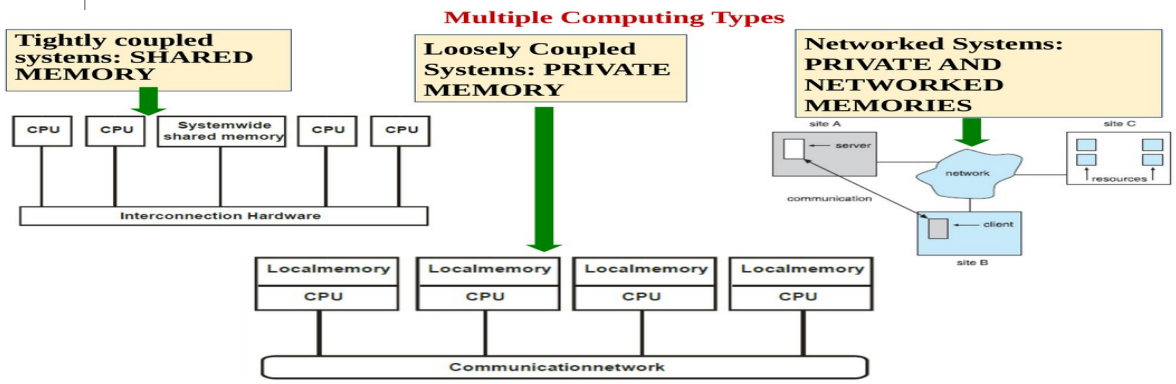
• They may be functionally quite dependent, and as a result take on the characteristics of a single node, a single 'system' rather than an interconnection of two nodes.

9. Differentiate between Loosely Coupled and Tightly Coupled Multiprocessor Systems

A:

S. O	LOOSELY COUPLED	TIGHTLY COUPLED
1.	There is distributed memory in loosely coupled multiprocessor system.	There is shared memory, in tightly coupled multiprocessor system.
2.	Loosely Coupled Multiprocessor System has low data rate.	Tightly coupled multiprocessor system has high data rate.
3.	In loosely coupled multiprocessor, Memory conflicts don't take place.	While tightly coupled multiprocessor system have memory conflicts.
4.	Loosely Coupled Multiprocessor system has low degree of interaction between tasks.	Tightly Coupled multiprocessor system has high degree of interaction between tasks.
5.	In loosely coupled multiprocessor, there is direct connection between processor and I/O devices.	While in tightly coupled multiprocessors, IOPIN helps connection between processor and I/O

		devices.
6.	Applications of loosely coupled multiprocessors are in DCSs.	Applications of tightly coupled multiprocessors are in parallel processing systems.



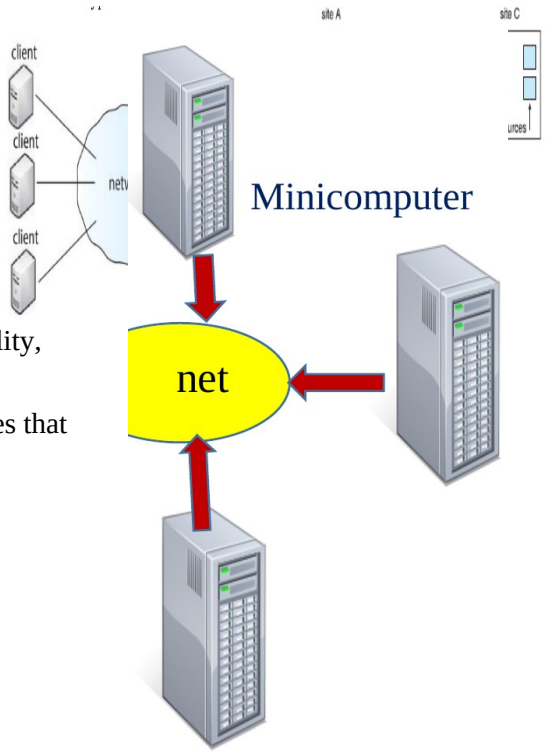
10. Explain networking distributed Multiprocessor System?

A: • There are many types of networks displaying a broad range of characteristics.

• They can vary from completely autonomous systems with the added ability to communicate electronically, to diskless workstations completely dependent on the network server for all secondary storage and communications operations.

• Each type of network has some unique set of properties that makes it attractive based on service provided, cost, reliability, serviceability, flexibility, etc.

• As well, each unique network possesses some set of properties that make it similar in nature to other network types.



• loosely coupled systems using networks are referred to as DCS. In these systems, the processors do not share memory, and each processor has its own local memory.

• If a processor writes the value 100 to the memory location x, this write operation will only change the contents of its local memory and will not affect the contents of the memory.

• All physical communication between the processors is done by passing messages across the interconnection network.

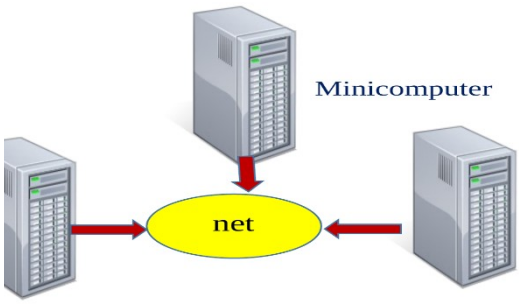
• Processors can be located far from each other to cover a wider geographical area, DCS are more freely expandable and can have an almost unlimited number of processors.

• DCS is basically a collection of processors interconnected by a communication network in which each processor has its own local memory and other peripherals, and the communication between any two processors of the system takes place by message passing over the communication network.

• For a particular processor, its own resources are local, whereas the other processors and their resources are remote. A processor and its resources are referred to as a node or site or machine of the DCS.

11. Explain minicomputer distributed Multiprocessor System?

A: The minicomputer model is a simple extension of the centralized time-sharing system.



• A DCS consists of a few minicomputers interconnected by a communication network where each minicomputer usually has multiple users simultaneously logged on to it.

• Several interactive terminals are connected to each minicomputer. Each user logged on to one specific minicomputer has remote access to other minicomputers.

• The network allows a user to access remote resources that are available on some machines other than the one on which the user is currently logged.

• The minicomputer model may be used when resource sharing with remote users is desired.

• Early Internet is example of minicomputer DCS.

Drawbacks of Minicomputer Model:

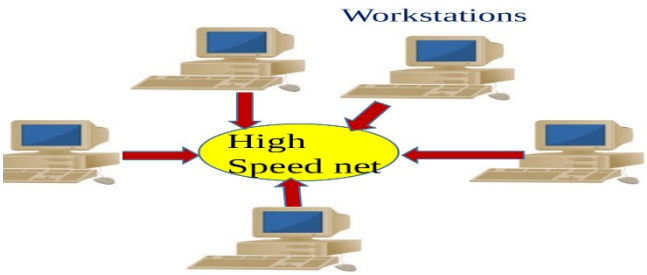
• Each user is logged on to one specific minicomputer, with remote access to other minicomputers.

• The processing of tasks is done by the processors of a single minicomputer.

• This model is very expensive.

12. Explain workstation distributed Multiprocessor System?

A: • WS Model Consists of network of powerful PCs, Each one with its own hard disk and local filesystem, Interconnected over high speed network



• A DCS based on the WS model consists of several WSs interconnected by a communication network.

• An organization may have several WSs located throughout an infrastructure where each WS is equipped with its own disk & serves as a single-user computer.

• In such an environment, at any one time a significant proportion of the WSs are idle which results in the waste of large amounts of CPU time.

• Therefore, the idea of the WS model is to interconnect all these WSs by a high-speed LAN so that idle WSs used to process jobs of users who are logged onto other WSs & do not have sufficient processing power at their own WSs to get their jobs processed efficiently.

• Example: Sprite system & Xerox PARC

13. Explain workstation server distributed Multiprocessor System?

A: • WS-Server Model Consists of multiple workstations coupled with powerful servers with extra hardware to store the file systems and other software like databases



• The WS model is a network of DISK-FULL PC WS having its own disk & a local file system.

• A DCS based on the WS-server model consists of a few minicomputers and several WSs interconnected by a network.

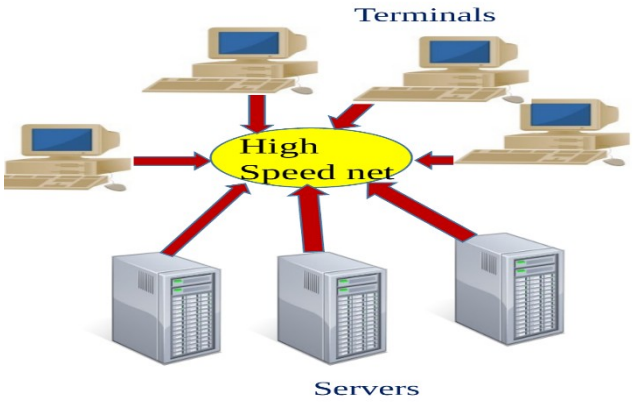
• A user logs onto a WS called his own PC.

• Normal computation activities required by the user's processes are performed at the user's PC, but requests for services provided by special servers are sent to a server providing that type of service that performs the user's requested activity & returns the result of request processing to the user's workstation.

• The user's processes need not migrated to the server machines for getting the work done by those machines.

14. Explain Multiprocessor pool distributed Systems?

A: • Consists of multiple processors: a pool of processors and a group of workstations



• The processor-pool model is based on the observation that most of the time a user does not need any computing power but once in a while the user may need a very large amount of computing power for a short time.

• Therefore, unlike the WS-SERVER model in which a processor is allocated to each user, in processor-pool model the processors are pooled together to be shared by the users as needed.

• The pool of processors consists of a large number of microcomputers & minicomputers attached to the network.

• Each processor in the pool has its own memory to load & run a system program or an application program of the distributed computing system.

• In this model no home machine is present & the user does not log onto any machine.

• This model has better utilization of processing power & greater flexibility.

• Example: Cambridge Distributed Computing System.

********** G O O D L U C K**********